trd.md

> **CivicSync · Technical Requirements Document** Version 1.0-rc1 · 28 Oct 2026 · Owner: Backend Lead (Manindra Shukla) · Co-owner: CV Lead (Aayush) Numbers here are **targets**, not measurements. Tunable business numbers live in `rules.md`; this file points to them rather than repeating them wherever it can.

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1. What "done" means technically

A phone with no signal can take a photo, name the hazard, work out a starting severity and save a report. When the signal returns, the report reaches a Postgres database, is placed in the right ward, is grouped with nearby repeats, gets a priority score, and shows up in a ranked queue that an officer can act on. The citizen can see each status change.

The proposal's flow is steps 1–8: Capture, Locate, Classify, Score, Submit, Deduplicate, Act, Notify. Steps 1–4 run on the phone, 5–6 in the cloud, 7–8 in the municipal workflow. This document keeps that numbering.

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2. Performance targets

Reference device **RD-1** (**A-04**): a mid-range Android phone with 4 GB RAM and an octa-core CPU of roughly the Snapdragon 600/700 or Helio G-series class, Android 11 or later, bought in the ₹10,000–15,000 range. We measure on at least two real devices; the slowest one counts.

ID	Requirement	Target	Notes
TRD-P1	On-device inference, one image, after warm-up	**< 150 ms (p95)**	From the proposal. CPU with XNNPACK first; GPU/NNAPI delegates only if they help on RD-1.
TRD-P2	Model file size	⩽ 5 MB	The deck estimated ~4 MB (MobileNetV2-class, INT8).
TRD-P3	Capture to "Thanks" on 4G	**< 3 s**	Uses the compressed photo (target ~200 KB). Upload time is expected to be well under this; the deck's estimate is ~0.3 s on 4G.
TRD-P4	User time from opening the app to "Thanks"	⩽ 30 s for 80% of first-time users	This is the survey's "under 30 seconds" test (PRD-M1). Different from TRD-P3: this includes the person's own taps.
TRD-P5	App cold start to live camera	⩽ 3 s on RD-1	
TRD-P6	Server-side grouping and scoring after a report lands	⩽ 2 s	One database function call.
TRD-P7	Dashboard queue first paint, 1,000 open tickets	⩽ 2 s on office broadband	Paged, 50 per page.
TRD-P8	Status-change push to citizen	⩽ 60 s (online phones)	Firebase Cloud Messaging.
TRD-P9	Photo size after compression	Long edge ⩽ 1280 px, WebP, target ⩽ 200 KB, hard cap 300 KB	A 320 px thumbnail ⩽ 30 KB is also uploaded.
TRD-P10	Expected load for design	500 reports/day average (~0.006 writes/s); 100× burst ≈ 0.6 writes/s	From the deck's calculation. The *beta* is expected to be far smaller.

2.1 Storage budget: the real bottleneck

The deck calculates 500 reports/day × ~200 KB ≈ 100 MB/day, or about 3 GB/month. Supabase's free plan, as of the 2026 pricing pages we checked, includes roughly 1 GB of file storage and 500 MB of database, and pauses projects after a week of inactivity (**C-08**; verify on supabase.com/pricing before committing). So:

- Before Sprint 5 beta we move to the **Pro plan (~$25/month)**, which has a much larger storage allowance and no pause.
- Retention (`rules.md` §7) deletes full photos of resolved tickets on a schedule.
- A daily storage-use alert fires at 70% of the allowance.

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3. Offline behaviour

ID	Requirement
TRD-O1	Steps 1–4 (capture, locate, classify, score) work with **no network**. The model ships inside the app.
TRD-O2	Unsent reports are kept in a local database for up to **7 days** and up to **200 reports**. After 7 days, the person is asked whether to send or discard.
TRD-O3	Each report gets a client-made UUID (`client_report_id`). Retried uploads use the same ID, so the server ignores copies.
TRD-O4	Sync runs when the app opens, when connectivity returns, and in a background task (Android WorkManager). Retries back off: 30 s, 2 min, 10 min, 1 h, then daily.
TRD-O5	The photo uploads first, then the report record is created. If the second step fails, the next retry reuses the uploaded photo.
TRD-O6	"My reports" shows queued items with a clear "Waiting to send" label.
TRD-O7	The capture time and GPS reading are taken when the photo is taken, not when it is sent. Reports older than 72 hours at send time are flagged `stale_capture` for officer attention (`rules.md` §2).
TRD-O8	The map in the app shows a plain pin with no street map when offline. We do not promise offline map tiles in v1.
TRD-O9	The dashboard needs a connection. It shows a banner when offline and does not cache sensitive photos.

4. Data model

PostgreSQL with the PostGIS extension, hosted on Supabase (**D-02**). Names are in `snake_case`. The proposal's ticket fields (`id, class, severity, confidence, lat, lng, ward, timestamp, image_url, status`) are split between *reports* and *tickets* so that duplicates make sense (**D-03**):

- A **report** is one citizen submission.
- A **ticket** is one real-world hazard; it holds one or more reports.

4.1 Tables

`cities` — `id`, `name`, `state`, `is_pilot`.

`wards` — `id`, `city_id`, `code`, `name`, `geom geometry(MultiPolygon, 4326)`. GiST index on `geom`. Source of boundaries is **Q-04**.

`profiles` — `id` (equals Auth user id), `role` (enum: `citizen`, `ward_officer`, `crew_lead`, `zonal_engineer`, `city_admin`, `platform_admin`), `display_name`, `ward_ids int[]`, `is_active`, `created_at`. Citizens start anonymous; no phone number or email is required.

`reports`

Column	Type	Meaning
`id`	uuid	Server id
`client_report_id`	uuid, unique	Idempotency key from the phone
`ticket_id`	uuid → tickets	The ticket this report belongs to
`reporter_id`	uuid → profiles	Pseudonymous account
`class_model`	text	What the model suggested
`class_final`	text	What the person confirmed (`pothole` , `open_drain` , `broken_road` , `waterlogging` , `other`)
`model_confidence`	real 0–1	Model's confidence in `class_model`
`extent`	text	`small` , `medium` , `large` , `not_sure`
`severity_prelim`	smallint 0–100	Phone's starting severity (`rules.md` §1.5)
`geom`	geography(Point, 4326)	Where the photo was taken
`gps_accuracy_m`	real	Reported accuracy
`captured_at`	timestamptz	When the photo was taken
`received_at`	timestamptz	When the server got it
`ward_id`	int → wards	From point-in-polygon
`photo_path` , `thumb_path`	text	Storage paths (private bucket)
`photo_sha256`	text	Exact-copy check
`quality_flags`	text[]	e.g. `blurry` , `stale_capture` , `low_gps` , `mock_location` , `privacy_blur_failed`
`note`	text, ⩽ 200 chars	Optional (used for "Other")

Column	Type	Meaning
`app_version`, `model_version`	text	For debugging and the learning loop

`tickets`

Column	Type	Meaning
`id`	uuid	
`class`	text	Same five values as `class_final`
`geom`	geography(Point, 4326)	Location of the first report; recomputed as the median point as reports join
`ward_id`	int	
`status`	enum	`reported`, `verified`, `assigned`, `resolved`, `rejected`
`priority_score`	smallint 0–100	`rules.md` §1
`priority_band`	enum	`critical`, `high`, `medium`, `low`
`location_context`	real 0–1	Output of the context lookup
`report_count`	int	All reports
`reporter_count`	int	Distinct reporters
`reopen_count`	int	
`first_reported_at`, `verified_at`, `assigned_at`, `resolved_at`	timestamptz	
`assigned_to`	uuid	A Crew Lead
`stage_due_at`	timestamptz	Deadline for the current stage (`rules.md` §4)
`escalation_level`	smallint 0–3	
`resolution_photo_path`	text	"After" photo
`reject_reason`	text	From a fixed list (`rules.md` §3.4)

`ticket_events` — `id`, `ticket_id`, `event_type` (`created`, `report_added`, `status_changed`, `merged`, `split`, `escalated`, `reopened`, `note`), `from_status`, `to_status`, `actor_id`, `note`, `created_at`. Append-only; this is the audit trail and the citizen's timeline.

`duplicate_candidates` — `id`, `ticket_a`, `ticket_b`, `distance_m`, `state` (`open`, `merged`, `kept_apart`), `decided_by`, `decided_at`.

`context_features` — `id`, `kind` (`school`, `hospital`, `market`, `bus_stand`, `rail_station`, `road_primary`, `road_secondary`, `road_local`), `geom`, `source` (e.g. `osm`), `imported_at`. Used to compute `location_context`.

`rule_config` — `key`, `value jsonb`, `version`, `updated_by`, `updated_at`. Holds the weights, thresholds and SLA numbers from `rules.md` so they can be tuned in Sprint 5 without a new release.

`device_tokens` — `user_id`, `fcm_token`, `platform`, `updated_at`.

`audit_log` — staff sign-ins, exports, role changes, photo views by staff.

4.2 Indexes and rules of the road

- GiST on `reports.geom`, `tickets.geom`, `wards.geom`, `context_features.geom`.
- B-tree on `tickets(status, priority_score desc)` and `tickets(ward_id, status)`.
- Row-level security on every table (`security.md` §4).
- All times stored in UTC; shown in IST.

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5. Interfaces

5.1 Phone → cloud

1. Upload photo and thumbnail to the private bucket `report-photos` under `reports/{user_id}/{client_report_id}`.
2. Call `submit_report(...)` (a Postgres function exposed as an RPC). It validates, assigns the ward, groups duplicates, scores, and returns the ticket id and status.
3. `get_nearby_open(lat, lng, radius_m)` returns open hazards nearby with class, status and report count, **no photos**.
4. `my_reports()` returns the citizen's own reports with ticket status and timeline.

5.2 Dashboard → cloud

`list_tickets(filters, page)`, `ticket_detail(id)`, `transition_ticket(id, to_status, note, resolution_photo?)`, `merge_tickets(a, b)`, `split_ticket(id, report_ids)`,

`list_overdue()`, `heat_cells(bbox, cell_m)` (grid-binned counts used for the density view, **D-07**).

5.3 External services

Service	Used for	Constraint
Supabase (Auth, Postgres/PostGIS, Storage, Edge Functions, Realtime, pg_cron)	Core back end	Free plan pauses after 1 week idle; Pro ≈ $25/month.
Firebase Cloud Messaging, Crashlytics, App Distribution	Push, crash reports, beta APKs	Free.
Google Maps Platform (map display in app and dashboard; optional reverse geocoding for a text label)	Maps	Each billable SKU has its own monthly free allowance (Essentials about 10,000 events). Set quotas and budget alerts. Do not store geocoding results long-term.
OpenStreetMap data (via Overpass or an extract)	Roads, schools, hospitals, markets for `context_features`	Open Database Licence: credit OSM and share derived database changes. Do not hit public servers for every request.
TensorFlow Lite / LiteRT runtime via `tflite_flutter`	On-device inference	Confirm package version works with the Flutter version in use.
Google ML Kit (on-device face detection, text recognition)	Privacy blur	Runs on the phone; no data sent to Google for these on-device modes.

Ward lookup uses our own PostGIS polygons, not Google's geocoder (**D-08**). It's cheaper, works inside the free allowance, and avoids storing Google results.

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6. AI requirements

ID	Requirement
TRD-A1	Five output classes: `pothole`, `open_drain`, `broken_road`, `waterlogging`, `none_of_these`. UI maps `none_of_these` and low confidence to "Other / pick one".
TRD-A2	Input 224×224 RGB. Transfer learning from an ImageNet-pre-trained MobileNetV2-class backbone in Keras, then post-training INT8 quantization with a representative dataset.
TRD-A3	Quality targets: top-1 ⩾ 80% and macro-F1 ⩾ 0.75 on the held-out Agra/Jodhpur field set (**A-05**). If not met, switch to the three-class fallback in `rules.md` §8.4.
TRD-A4	Data targets: ⩾ 400 images per class (public + field), of which ⩾ 100 per class are field images kept out of training as a test set. The train/test split is by *location*, not by random image, so near-identical frames don't leak.
TRD-A5	Every model release carries a version, training-data hash, metrics file and the INT8 vs. float comparison. Models are loaded from app assets in v1; remote model updates are out of scope.
TRD-A6	The model's guess never overrides the person's choice; see `rules.md` §8.
TRD-A7	Extent of damage in v1 comes from the person's size chip, because a plain classifier cannot measure size (**D-09**). A detection or segmentation model is a post-pilot upgrade.

7. Non-functional requirements

Area	Requirement
Platforms	Android 8.0 (API 26) or newer for v1 (**A-09**). iOS build is a stretch goal. Dashboard works on current Chrome, Edge and Firefox on a 1366×768 laptop.
App size	APK ⩽ 40 MB including the model (target).
Data use	⩽ 300 KB per report including thumbnail; no background data use beyond sync and push.
Battery	No continuous GPS or camera use; location is read once per capture with a short timeout (10 s), falling back to the last known position flagged `low_gps`.
Reliability	Pilot target: best-effort 99% uptime outside provider incidents. No data loss for queued reports.
Scalability	Designed for one pilot city pair (Agra and Jodhpur); adding a city is a config and ward-import task, not code.
Observability	Crashlytics on the app; error and slow-query alerts on the database; a daily health summary to the team channel.
Maintainability	Business rules are data (`rule_config`), not hard-coded. Logic units covered by automated tests (target ⩾ 70% line coverage on rules and scoring code). Row-level-security policies covered by database tests.
Accessibility	Tap targets ⩾ 48 dp; text contrast to WCAG 2.1 AA; supports system font scaling to 200%; no information by colour alone (`design.md` §2).
Localisation	English and Hindi strings from day one in the string files, so Hindi can ship in Sprint 4 (FR-C8).
Cost	Total pilot cloud cost < $50/month (`tooling.md` §6).
Privacy and security	See `security.md`. In short: private photo bucket, short-lived links, role-based access, retention schedule.
Licensing	Prefer permissive licences (MIT, BSD, Apache-2.0). Flag copyleft (AGPL/GPL) dependencies in `tooling.md` before use.

8. Test and measurement requirements

What	How	Gate
Inference time	Benchmark script on RD-1 and one lower-end phone, 200 runs, report p50/p95	M3
Classifier quality	Held-out field set; confusion matrix; per-class precision/recall	M3, M5
End-to-end latency	Timed run on a real 4G SIM, 20 reports	M2, M5
Offline sync	Airplane-mode test: 10 queued reports, reconnect, verify none lost or doubled	M2
Duplicate grouping	Scripted test cases of near/far/class-mismatch points at and around the thresholds	M4
Row-level security	Database tests: a citizen can't read another's reports; an officer can't see another ward	M2, M4
Load	A scripted burst at 100× average (0.6 writes/s) for 10 minutes	M4
Usability	5 first-time users per city, timed, with notes	Sprint 5

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9. Document acceptance criteria

This document passes review when it meets the checklist for `trd.md` in `loop.md` §3.